

nnmclub-login- diagnosis-20260616

NNMClub Login Failure — Root-Cause Diagnosis

Revision: 1 **Last modified:** 2026-06-16T00:00:00Z **Run-id:** nnmclub-login-diagnosis-20260616 **Authority:** §11.4.6 (no-guessing, FACT-grade), §11.4.114 (working-example comparison), §11.4.123 (rock-solid proof) **Scope:** READ-ONLY investigation. No production edits, no commit. No credential values captured/printed.

1. Verdict (FACT-grade)

Root cause: nnmclub.to gates login.php behind a Cloudflare Turnstile JS challenge. The Boba password-login POST (`_nnmclub_login` in `download-proxy/src/merge_service/search.py`, and the plugin fallback `_password_login` in `plugins/nnmclub.py`) sends only `username / password / login`. The server requires a valid `cf-turnstile-response` token (produced by solving the JS widget in a real browser) before it will issue the `phpbb2mysql_4_sid` session cookie. Without it the POST returns **HTTP 200 with NO Set-Cookie** → no session → `has_session=False`.

Classification: (a) CAPTCHA → operator must supply browser-exported NNMCLUB_COOKIES. This is NOT a code bug in the POST body, NOT a UA/Cloudflare-block on requests (plain curl reaches the form fine), and NOT (provably) an invalid-credentials problem — credentials never get a chance to be evaluated because the Turnstile gate fails first. The dead-domain fix (`f6b2761`, `nnm-club.me`→`nnmclub.to`) correctly restored reachability; the remaining failure is the Turnstile login gate.

2. Captured evidence (from nezha.local — direct internet, no Mac SSH tunnel)

All probes run as `ssh milosvasic@nezha.local 'curl ...'`.

2a. Login form requires a JS CAPTCHA + hidden fields

GET `https://nnmclub.to/forum/login.php:`

```
<form action="login.php" method="post" target="_top"
id="loginFrm">
<input name="f"          type="hidden" value="-1">
<input                  type="hidden" name="redirect" value="" />
<input                  type="hidden" name="code"
value="5cab14e06b8e112b" />
... CAPTCHA ...
<noscript><b>Для отображение CAPTCHA, пожалуйста, включите
JavaScript</b></noscript>
```

CAPTCHA widget tokens present in the page:

```
cf-turnstile
turnstile
data-sitekey
```

→ **Cloudflare Turnstile.** The visible CAPTCHA only renders with JavaScript; a non-browser client cannot produce the `cf-turnstile-response` token.

2b. The hidden code field is a STATIC anti-CSRF token, not the blocker

Two separate GETs returned the **identical** value `code=5cab14e06b8e112b`. A per-session nonce would differ. So including code would not fix login — Turnstile is the gate.

2c. Credential POST (with hidden code included) yields NO session cookie

```
fresh code=5cab14e06b8e112b
POST .../login.php (username=probe_xyz, password=probe_xyz,
login=вход, redirect=, code=<above>)
HTTP/2 200
# (no Set-Cookie line in response headers)
```

A successful login would carry Set-Cookie: phpbb2mysql_4_sid=....
It is absent → server rejected the login at the Turnstile gate before evaluating credentials.

2d. Cloudflare in front, but NOT blocking the request

```
server: cloudflare
cf-cache-status: DYNAMIC
HTTP/2 200
```

The page loads (HTTP 200), so this is **not** a UA/IP Cloudflare challenge on the request itself — option (c) is ruled out. The block is the in-page Turnstile widget the login flow must satisfy.

3. Working-tracker comparison (§11.4.114)

Tracker	login.php Turnstile?	Boba auth method	has_session
rutracker.org	none (probe returned empty)	password POST (login_username/login_password/login)	True
kinozal	none observed	password POST	True
iptorrents	n/a (cookies)	—	True
nnmclub.to	cf-turnstile present	password POST (username/password/login) — Turnstile token missing	False

The Boba _nnmclub_login is a faithful clone of the rutracker pattern. It works for rutracker precisely because rutracker's login.php has **no Turnstile**. nnmclub's does, so the identical pattern cannot succeed there. The code is not “wrong” — the target's auth surface is fundamentally different (browser-only).

Probe contrast:

```
rutracker login.php turnstile grep → (empty)    # no widget
nnmclub  login.php turnstile grep → cf-turnstile / turnstile /
data-sitekey
```

4. Why the existing diag message is misleading (§11.4.6)

search.py:1538-1547 records on failure:

```
"nmclub login returned no session cookie – likely credential failure"
```

The word “**likely**” is a §11.4.6 no-guessing violation, and the guess is wrong: the FACT is a Turnstile gate, not a credential failure. Suggested corrected wording:

```
"nmclub login.php is gated by Cloudflare Turnstile (JS CAPTCHA);  
password POST cannot obtain a session – set NNMCLUB_COOKIES from a  
browser session."
```

5. Classification & required action

Classification (a): CAPTCHA → browser-exported cookies (operator/env, §11.4.3 topology).

Operator action (the working path — already supported by code)

_search_nmclub (search.py:1466-1474) ALREADY prefers NNMCLUB_COOKIES when set and skips the password login entirely. The operator must export cookies from a logged-in browser session at nmclub.to and set the NNMCLUB_COOKIES env var.

Exact cookies required (the code’s hard gate is phpbb2mysql_4_sid; a full working session also carries the data + persistence cookies — export all phpbb2mysql_4_*): -
phpbb2mysql_4_sid ← **REQUIRED** — search.py:1479 / plugin login() line 212 reject without it - phpbb2mysql_4_data ← needed for persistent auth (plugin_init line 341 keys on this) - cf_clearance ← recommended — Cloudflare clearance cookie, lets subsequent requests bypass the Turnstile interstitial

NNMCLUB_COOKIES format (semicolon-separated, per search.py:1470 and plugin login() line 188):

```
NNMCLUB_COOKIES="phpbb2mysql_4_sid=<...>;  
phpbb2mysql_4_data=<...>; cf_clearance=<...>"
```

Cookies expire periodically (same operational caveat as RuTracker per CLAUDE.md) and must be re-exported when nmclub auth lapses.

NOT recommended (out of scope / fragile)

Automating Turnstile solving (3rd-party solver services, headless-browser token mint) is brittle, may violate the tracker ToS, and is not the established Boba pattern. The cookie-injection path (§11.4.3 operator-supplied evidence) is the correct, robust fix.

No code fix needed for login to work

The cookie path already works end-to-end in code. The only code-quality follow-ups (separate, optional): 1. Fix the §11.4.6 “likely credential failure” diag string (section 4) → state the Turnstile FACT so operators know to supply cookies. 2. Optionally have `_nnmclub_login` detect `cf-turnstile` in the login page and emit a `error_type: "upstream_captcha"` diag (mirroring `rutracker's upstream_captcha` label at `search.py:1255-1257`) instead of `auth_failure`.

RED-test design (for the diag-string follow-up, if pursued)

- RED: assert that when `_nnmclub_login` is driven against a login page containing `cf-turnstile`, the recorded diag `error_type == "upstream_captcha"` (currently `auth_failure`) and the message contains “Turnstile”, not “likely”. Fails pre-fix.
 - GREEN: classifier detects the `cf-turnstile` marker; flip `RED_MODE=0` as the guard.
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6. Files referenced

- `/Volumes/T7/Projects/boba/download-proxy/src/merge_service/search.py` — `_search_nnmclub` (1454), `_nnmclub_login` (1501), `diag` (1538), `cookie-precedence` (1466)
- `/Volumes/T7/Projects/boba/plugins/nnmclub.py` — `login()` (180), `_password_login()` (217), `_init()` cookie load (311)
- `/Volumes/T7/Projects/boba/download-proxy/src/api/auth.py` — `nnmclub auth_status` branch (not the cause; reflects `has_session` truthfully)